

UNIVERSITY OF COPENHAGEN

Lecture 1 - The model

The Merged Model Framework

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Introduction

Who am I?

Jonathan Wenzel Pedersen

- **Currently**: Research with Professor Finn Tarp on macroeconomic modelling and development-policy analysis
- **Previously**: Worked with developing large-scale macroeconomic forecasting models for the Danish Ministry of Finance

Viet Nam's 2026–2030 policy horizon

The 2026–2030 development plan sets an ambitious direction for the next five years:

- Average annual GDP growth of at least 10%
- GDP per capita of around USD 8,500 by 2030
- Viet Nam among the world's 30 largest economies by GDP by 2030

These goals define an ambitious policy agenda. Over the next days, we will use the Merged Model to assess their macroeconomic feasibility.

From growth targets to macroeconomic scenarios

A growth target is not yet a macroeconomic scenario.

- Growth must be supported by investment and imports
- The fiscal path must be feasible
- Credit, reserves, and foreign financing must remain consistent with the path

A macroeconomic scenario is credible only when these conditions can hold at the same time.

The Merged Model

A tool for turning policy ambitions into a coherent macroeconomic scenario.

- Starts from a proposed growth and policy path
- Shows what that path requires from the economy
- Highlights where the path may need to be adjusted

The model does not choose the target, it shows what it takes for the target to work.

Background of the Merged Model

The Merged Model was developed to bring development planning and macroeconomic stabilisation into one framework.

- Built on the World Bank's Revised Minimum Standard Model (RMSM)
- Combines RMSM with the IMF Financial Programming approach
- Originally developed by Brixen and Tarp, and later developed further by Jensen and Tarp

The result is a single framework for analysing growth, stabilisation, and financing together, that is consistent with the language and methodology of both the World Bank and IMF.

Why use this framework?

The Merged Model is designed to make ambitious policy scenarios easier to assess, discuss, and revise.

- **A feasibility tool:** tests whether an assumed growth and policy path can hold together economically
- **Simple on purpose:** based on a limited set of relationships, so assumptions are clear, easy to challenge, and easy to update
- **An internationally familiar framework:**
  - built on the World Bank RMSM tradition
  - combines this with IMF Financial Programming

It provides a common framework for domestic scenario work and dialogue with international partners.

Today's lecture

1. Introduction
2. RMSM
3. FP
4. Merged Model

RMSM

RMSM: Condensed Equation System

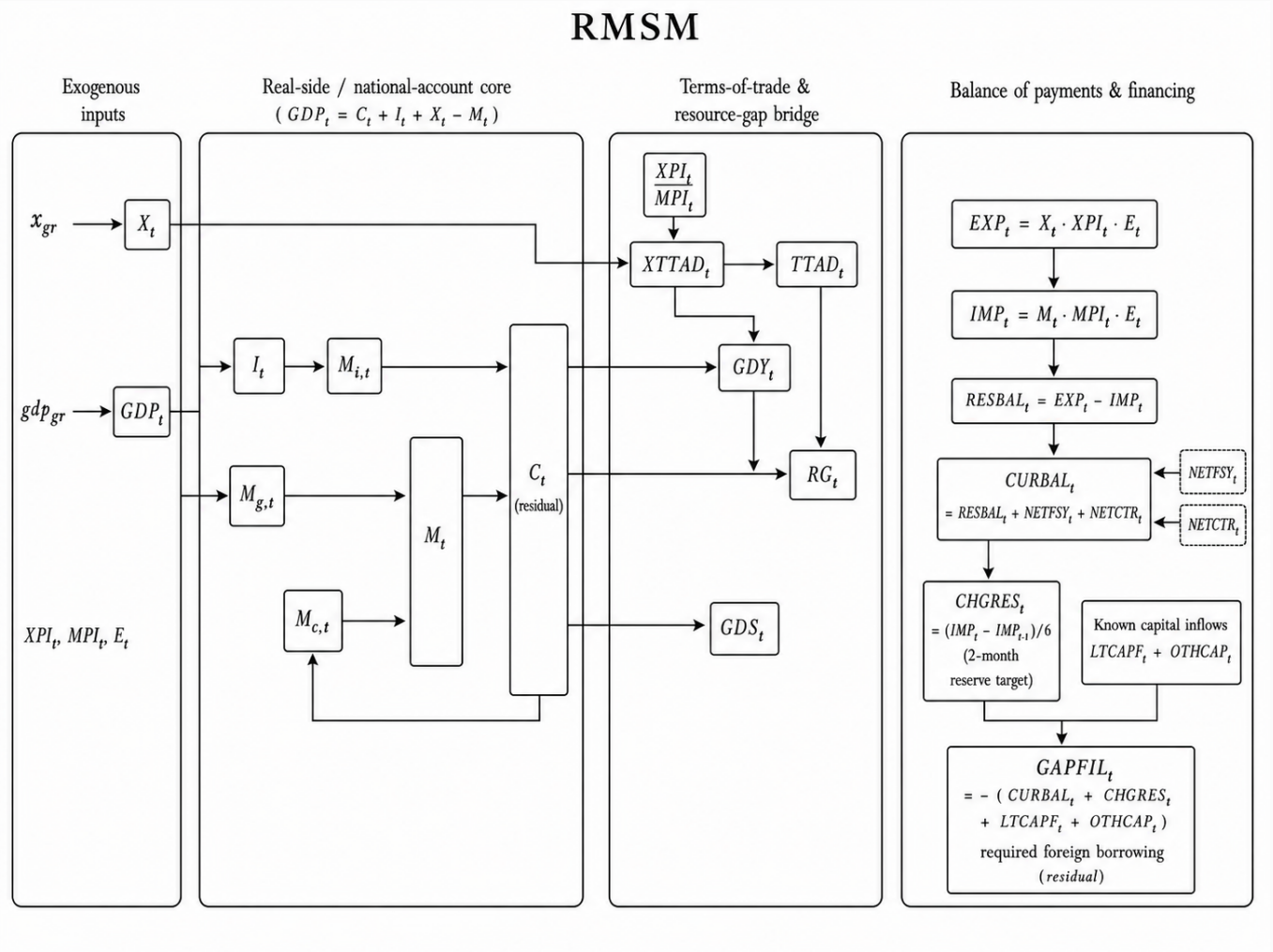
RMSM — National Accounts

	Equation	No.
Exogenous		
GDP	$GDP_t = GDP_{t-1}\left(1 + gdp_gr_t\right)$	(1)
Exports	$X_t = X_{t-1}\left(1 + x_gr_t\right)$	(2)
Endogenous		
Investment	$I_t = k_t + (A_{1,t} + A_{2,t})GDP_t - A_{2,t}GDP_{t-1}$	(3)
Imports		
Intermediate imports	$M_{g,t} = M_{g,t-1}\left[1 + mg_el_t\left(\frac{GDP_t}{GDP_{t-1}} - 1\right)\right]$	(4)
Investment imports	$M_{i,t} = M_{i,t-1}\left[1 + mi_el_t\left(\frac{I_t}{I_{t-1}} - 1\right)\right]$	(5)
Consumer imports	$M_{c,t} = M_{c,t-1}\left[1 + mc_el_t\left(\frac{C_t}{C_{t-1}} - 1\right)\right]$	(6)
Total imports	$M_t = M_{g,t} + M_{i,t} + M_{c,t}$	(7)
Terms-of-trade revaluations		
Terms-of-trade adjustment	$TTADJ_t = \frac{X_t XPI_t}{MPI_t} - X_t$	(8)
Gross domestic income	$GDY_t = GDP_t + TTADJ_t$	(9)
Export purchasing power	$XTTADJ_t = X_t + TTADJ_t$	(10)
Resource gap	$RG_t = M_t - XTTADJ_t$	(11)
Gross domestic saving	$GDS_t = I_t - RG_t$	(12)
Closure		
Consumption	$C_t = GDP_t - I_t - X_t + M_{g,t} + M_{i,t} + M_{c,t}$	(13)
	$C_t = GDY_t + RG_t - I_t$	

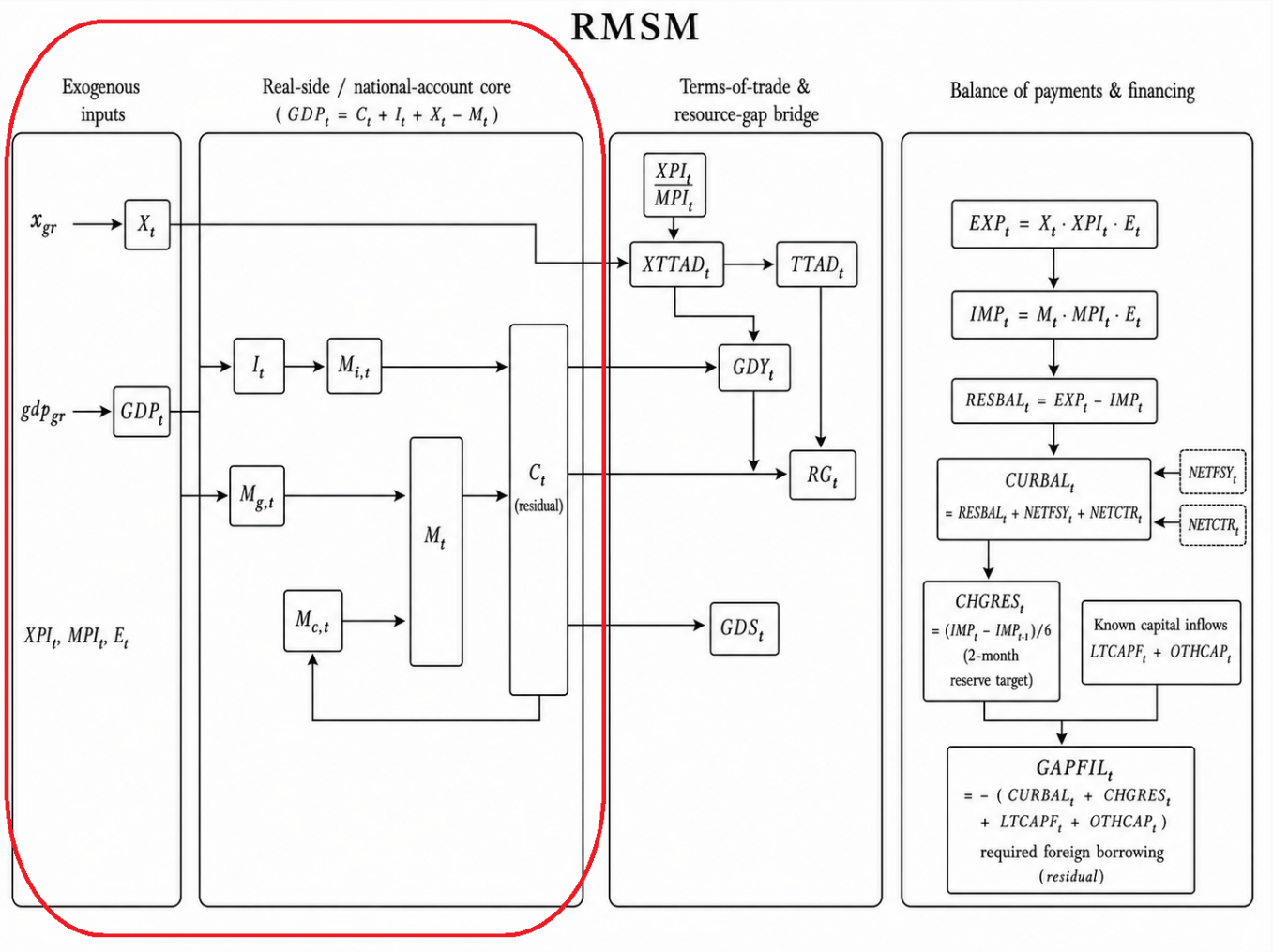
RMSM — Balance of Payments

	Equation	No.
Exogenous	$OTHFSY_t,\ NETCTR_t,\ LTCAPF_t,\ OTHCAP_t,\ i_t$	
Accounting identity		
Balance of payments	$CURBAL_t + NETCAP_t + CHGRES_t = 0$	(14)
Current account		
Current-account balance	$CURBAL_t = RESBAL_t + NETFSY_t + NETCTR_t$	(15)
Resource balance	$RESBAL_t = EXP_t - IMP_t$	(16)
Net factor services	$NETFSY_t = INTGAP_t + OTHFSY_t$	(17)
Capital account		
Capital-account balance	$NETCAP_t = NETGAP_t + LTCAPF_t + OTHCAP_t$	(18)
Net gap-filling capital	$NETGAP_t = GAPFIL_t - AMTGAP_t$	(19)
Trade valuation		
Export receipts	$EXP_t = X_t XPI_t E_t$	(20)
Import payments	$IMP_t = M_t MPI_t E_t$	(21)
Reserves		
Change in reserves	$CHGRES_t = \frac{IMP_t - IMP_{t-1}}{6}$	(22)
Closure		
Required foreign borrowing	$GAPFIL_t = -CURBAL_t - AMTGAP_t - LTCAPF_t - OTHCAP_t - CHGRES_t$	(23)
Debt dynamics		
Amortisation	$AMTGAP_t = GAPFIL_{t-1}$	(24)
Outstanding gap-filling debt	$DODGAP_t = DODGAP_{t-1} + GAPFIL_t - AMTGAP_t$	(25)
Interest on gap-filling debt	$INTGAP_t = DODGAP_{t-1} i_t$	(26)

RMSM model


RMSM
Exogenous inputs
Real-side / national-account core ($GDP_t = C_t + I_t + X_t - M_t$)
Terms-of-trade & resource-gap bridge
Balance of payments & financing
x_{gr}
X_t
gdp_{gr}
GDP_t
XPI_t, MPI_t, E_t
I_t
$M_{i,t}$
$M_{g,t}$
$M_{c,t}$
M_t
C_t (residual)
$\frac{XPI_t}{MPI_t}$
$XTTAD_t$
$TTAD_t$
GDY_t
RG_t
GDS_t
$EXP_t = X_t \cdot XPI_t \cdot E_t$
$IMP_t = M_t \cdot MPI_t \cdot E_t$
$RESBAL_t = EXP_t - IMP_t$
$CURBAL_t$ = $RESBAL_t + NETFSY_t + NETCTR_t$
$NETFSY_t$
$NETCTR_t$
$CHGRES_t$ = $(IMP_t - IMP_{t-1})/6$ (2-month reserve target)
Known capital inflows $LTCAPF_t + OTHCAP_t$
$GAPFIL_t$ = $-$ ($CURBAL_t + CHGRES_t + LTCAPF_t + OTHCAP_t$) required foreign borrowing (residual)

RMSM model - National Account



RMSM: National-account identity

$$\underbrace{GDP_t}_{\text{Gross domestic product}} = \underbrace{C_t}_{\text{Consumption}} + \underbrace{I_t}_{\text{Investment}} + \underbrace{X_t}_{\text{Exports}} - \underbrace{M_t}_{\text{Imports}}$$

RMSM: GDP path

$$GDP_t = \underbrace{GDP_{t-1}}_{\text{Previous-period GDP}} \left(1 + \underbrace{gdp_{gr_t}}_{\text{Assumed GDP growth}} \right)$$

RMSM: Export path

$$X_t = \underbrace{X_{t-1}}_{\text{Previous-period exports}} \left(1 + \underbrace{x_{gr_t}}_{\text{Assumed export growth}} \right)$$

RMSM: Investment requirement

$$I_t = \underbrace{k_t}_{\text{Baseline investment}} + \underbrace{A_{1,t} GDP_t}_{\text{Investment related to GDP}} + \underbrace{A_{2,t} \left(GDP_t - GDP_{t-1} \right)}_{\text{Investment related to GDP growth}}$$

RMSM: Import requirements

$$\underbrace{M_t}_{\text{Total imports}} = \underbrace{M_{g,t}}_{\text{Intermediate imports}} + \underbrace{M_{i,t}}_{\text{Investment imports}} + \underbrace{M_{c,t}}_{\text{Consumer imports}}$$

$$M_{g,t} = M_{g,t-1}\left[1 + \underbrace{\varepsilon_{g,t}}_{\text{GDP elasticity}}\left(\frac{GDP_t}{GDP_{t-1}} - 1\right)\right]$$

$$M_{i,t} = M_{i,t-1}\left[1 + \underbrace{\varepsilon_{i,t}}_{\text{Investment elasticity}}\left(\frac{I_t}{I_{t-1}} - 1\right)\right]$$

$$M_{c,t} = M_{c,t-1}\left[1 + \underbrace{\varepsilon_{c,t}}_{\text{Consumption elasticity}}\left(\frac{C_t}{C_{t-1}} - 1\right)\right]$$

RMSM: Closing the national accounts

$$GDP_t = C_t + I_t + X_t - M_t$$

$$\Longrightarrow$$

$$\underbrace{C_t}_{\text{Consumption}} = GDP_t - I_t - X_t + M_t$$

$$= GDP_t - I_t - X_t + M_{g,t} + M_{i,t} + M_{c,t}$$

Since $M_{c,t}$ depends on C_t, consumption and consumer imports are solved jointly.

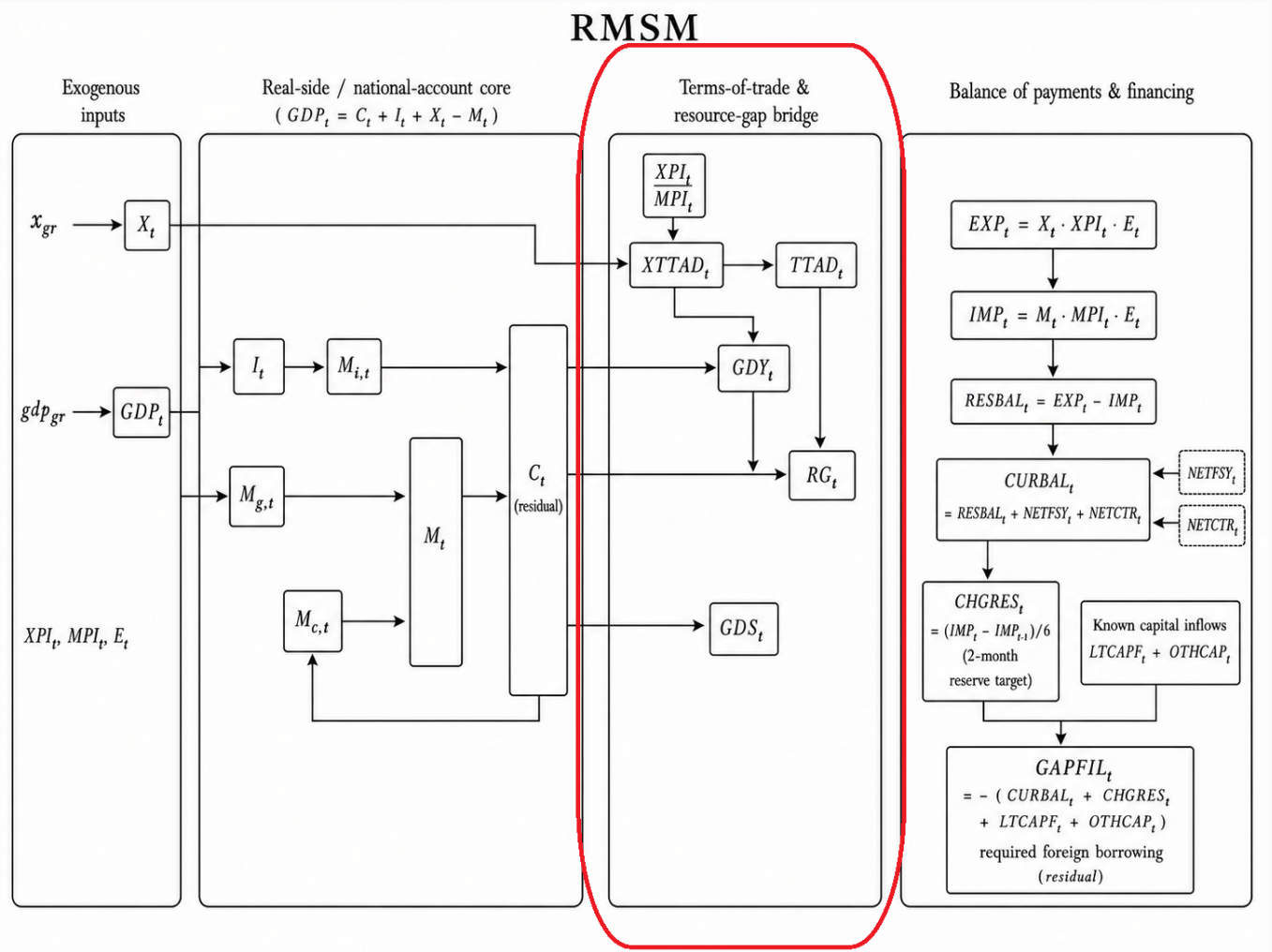
RMSM: National Accounts sum up

$$\underbrace{GDP_t,\ X_t}_{\text{scenario inputs}} \quad \Longrightarrow \quad \underbrace{I_t,\ M_{g,t},\ M_{i,t}}_{\text{growth requirements}}$$

$$\underbrace{C_t,\ M_{c,t}}_{\text{solved jointly}} \quad \Longrightarrow \quad \underbrace{M_t}_{\text{total imports}}$$

$$\Longrightarrow \quad \underbrace{GDP_t,\ C_t,\ I_t,\ X_t,\ M_t}_{\text{real-side projection}}$$

RMSM - Decomposition foreign real resources



RMSM: Export purchasing power

$$\underbrace{XTTADJ_t}_{\text{Export purchasing power}} = \underbrace{X_t}_{\text{Export volume}} \underbrace{\left(\frac{XPI_t}{MPI_t}\right)}_{\text{Terms of trade}}$$

$$\frac{XPI_t}{MPI_t} > 1 \implies XTTADJ_t > X_t$$

$$\frac{XPI_t}{MPI_t} < 1 \implies XTTADJ_t < X_t$$

RMSM: Terms-of-trade adjustment

$$\underbrace{TTADJ_t}_{\text{Terms-of-trade adjustment}} = \underbrace{XTTADJ_t}_{\text{Export purchasing power}} - \underbrace{X_t}_{\text{Export volume}}$$

$$\underbrace{GDY_t}_{\text{Gross domestic income}} = \underbrace{GDP_t}_{\text{Gross domestic product}} + \underbrace{TTADJ_t}_{\text{Terms-of-trade adjustment}}$$

RMSM: Resource gap

$$\underbrace{RG_t}_{\text{Resource gap}} = \underbrace{M_t}_{\text{Imports}} - \underbrace{XTTADJ_t}_{\text{Export purchasing power}}$$

$$RG_t > 0 \implies \text{Net real resources from abroad}$$

$$RG_t < 0 \implies \text{Net real resources supplied abroad}$$

RMSM: Domestic saving and investment

$$\underbrace{I_t}_{\text{Investment requirement}} = \underbrace{GDS_t}_{\text{Gross domestic saving}} + \underbrace{RG_t}_{\text{Foreign real resources}}$$

$$\underbrace{GDS_t}_{\text{Gross domestic saving}} = \underbrace{GDY_t}_{\text{Gross domestic income}} - \underbrace{C_t}_{\text{Consumption}}$$

$$GDS_t = I_t - RG_t$$

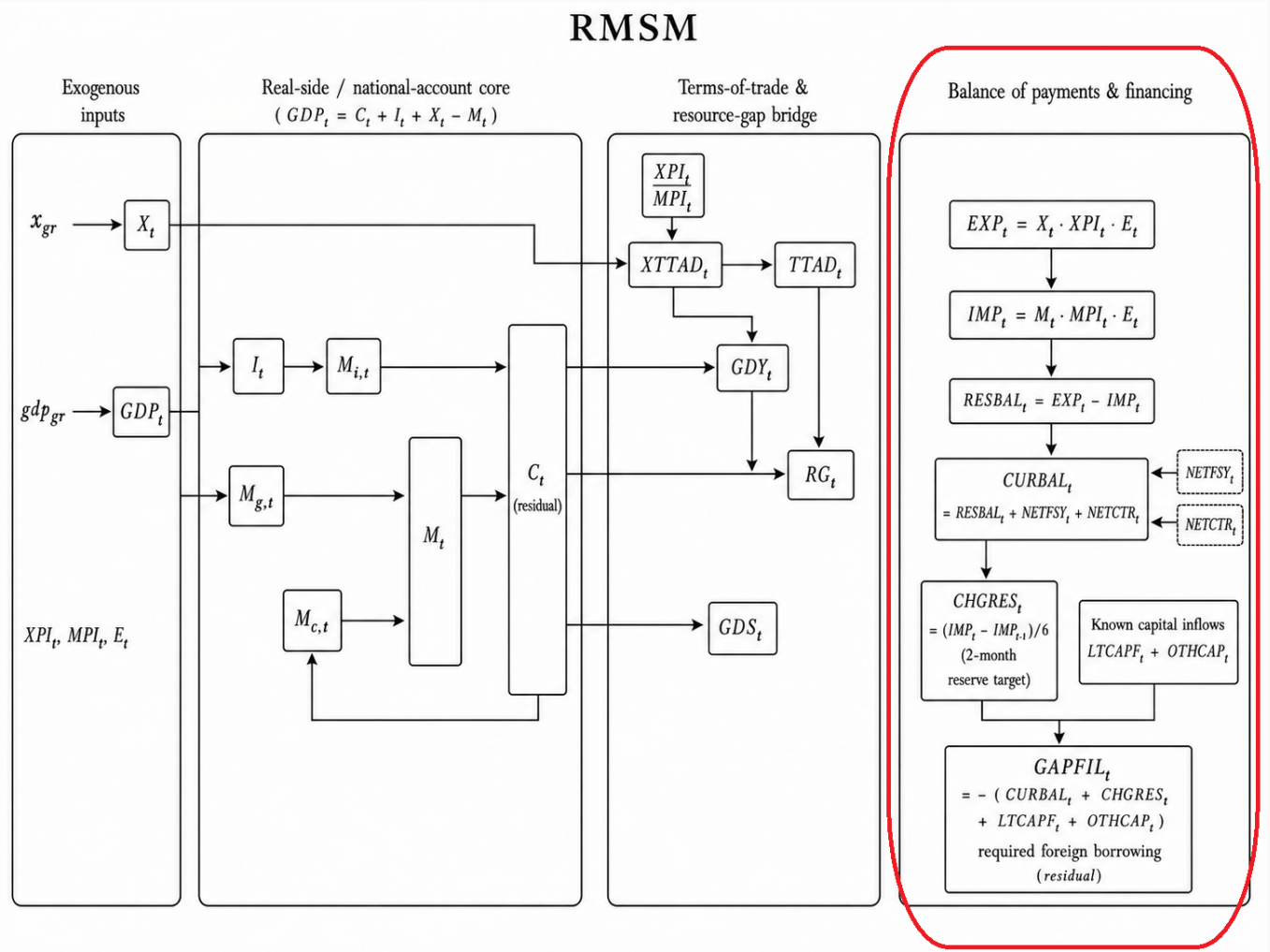
RMSM: Foreign real resources — summary

$$\underbrace{\frac{XPI_t}{MPI_t}}_{\text{Terms of trade}} \Longrightarrow \underbrace{XTTADJ_t}_{\text{Export purchasing power}}$$

$$RG_t = \underbrace{M_t - XTTADJ_t}_{\text{Resource gap}} \Longrightarrow \underbrace{\text{Net real resources from abroad}}_{\text{Real-resource view}}$$

$$\underbrace{I_t}_{\text{Investment requirement}} = \underbrace{GDS_t}_{\text{Domestic saving}} + \underbrace{RG_t}_{\text{Foreign real resources}}$$

RMSM - Balance of payments



RMSM: Valuing trade flows

$$\underbrace{EXP_t}_{\text{Export receipts}} = \underbrace{X_t}_{\text{Export volume}} \quad \underbrace{XPI_t}_{\text{Export price index}} \quad \underbrace{E_t}_{\text{Exchange rate}}$$

$$\underbrace{IMP_t}_{\text{Import payments}} = \underbrace{M_t}_{\text{Import volume}} \quad \underbrace{MPI_t}_{\text{Import price index}} \quad \underbrace{E_t}_{\text{Exchange rate}}$$

RMSM: Resource balance

$$\underbrace{RESBAL_t}_{\text{Resource balance}} = \underbrace{EXP_t}_{\text{Export receipts}} - \underbrace{IMP_t}_{\text{Import payments}}$$

$RESBAL_t > 0 \quad \Longrightarrow \quad$ Export receipts exceed import payments

$RESBAL_t < 0 \quad \Longrightarrow \quad$ Import payments exceed export receipts

RMSM: Net factor services and transfers

$$\underbrace{NETFSY_t}_{\text{Net factor services}} = \underbrace{INTGAP_t}_{\text{Interest on gap-filling debt}} + \underbrace{OTHFSY_t}_{\text{Other factor services}}$$

$$\underbrace{NETCTR_t}_{\text{Net current transfers}}$$

RMSM: Current-account balance

$$\underbrace{CURBAL_t}_{\text{Current-account balance}} = \underbrace{RESBAL_t}_{\text{Resource balance}} + \underbrace{NETFSY_t}_{\text{Net factor services}} + \underbrace{NETCTR_t}_{\text{Net current transfers}}$$

RMSM: Capital flows and reserve accumulation

$$\underbrace{NETCAP_t}_{\text{Capital-account balance}} = \underbrace{NETGAP_t}_{\text{Net gap-filling capital}} + \underbrace{LTCAPF_t}_{\text{Other long-term capital}} + \underbrace{OTHCAP_t}_{\text{Other capital}}$$

$$\underbrace{NETGAP_t}_{\text{Net gap-filling capital}} = \underbrace{GAPFIL_t}_{\text{New foreign borrowing}} - \underbrace{AMTGAP_t}_{\text{Amortisation}}$$

$$\underbrace{CHGRES_t}_{\text{Change in reserves}} = \frac{IMP_t - IMP_{t-1}}{6}$$

RMSM: Closing the balance of payments

$$\underbrace{CURBAL_t}_{\text{Current account}} + \underbrace{NETCAP_t}_{\text{Capital account}} + \underbrace{CHGRES_t}_{\text{Reserve accumulation}} = 0$$

$$\underbrace{GAPFIL_t}_{\text{Required foreign borrowing}} = -CURBAL_t - AMTGAP_t - LTCAPF_t - OTHCAP_t - CHGRES_t$$

RMSM: Gap-filling debt dynamics

$$\underbrace{AMTGAP_t}_{\text{Amortisation}} = \underbrace{GAPFIL_{t-1}}_{\text{Previous-period foreign borrowing}}$$

$$\underbrace{DODGAP_t}_{\text{Outstanding gap-filling debt}} = DODGAP_{t-1} + GAPFIL_t - AMTGAP_t$$

$$\underbrace{INTGAP_t}_{\text{Interest on gap-filling debt}} = \underbrace{DODGAP_{t-1}}_{\text{Previous-period debt}} \underbrace{i_t}_{\text{Interest rate}}$$

RMSM: External financing — summary

$$\underbrace{X_t,\ M_t}_{\text{Real trade flows}} \quad \Longrightarrow \quad \underbrace{EXP_t,\ IMP_t}_{\text{Foreign-currency trade flows}}$$

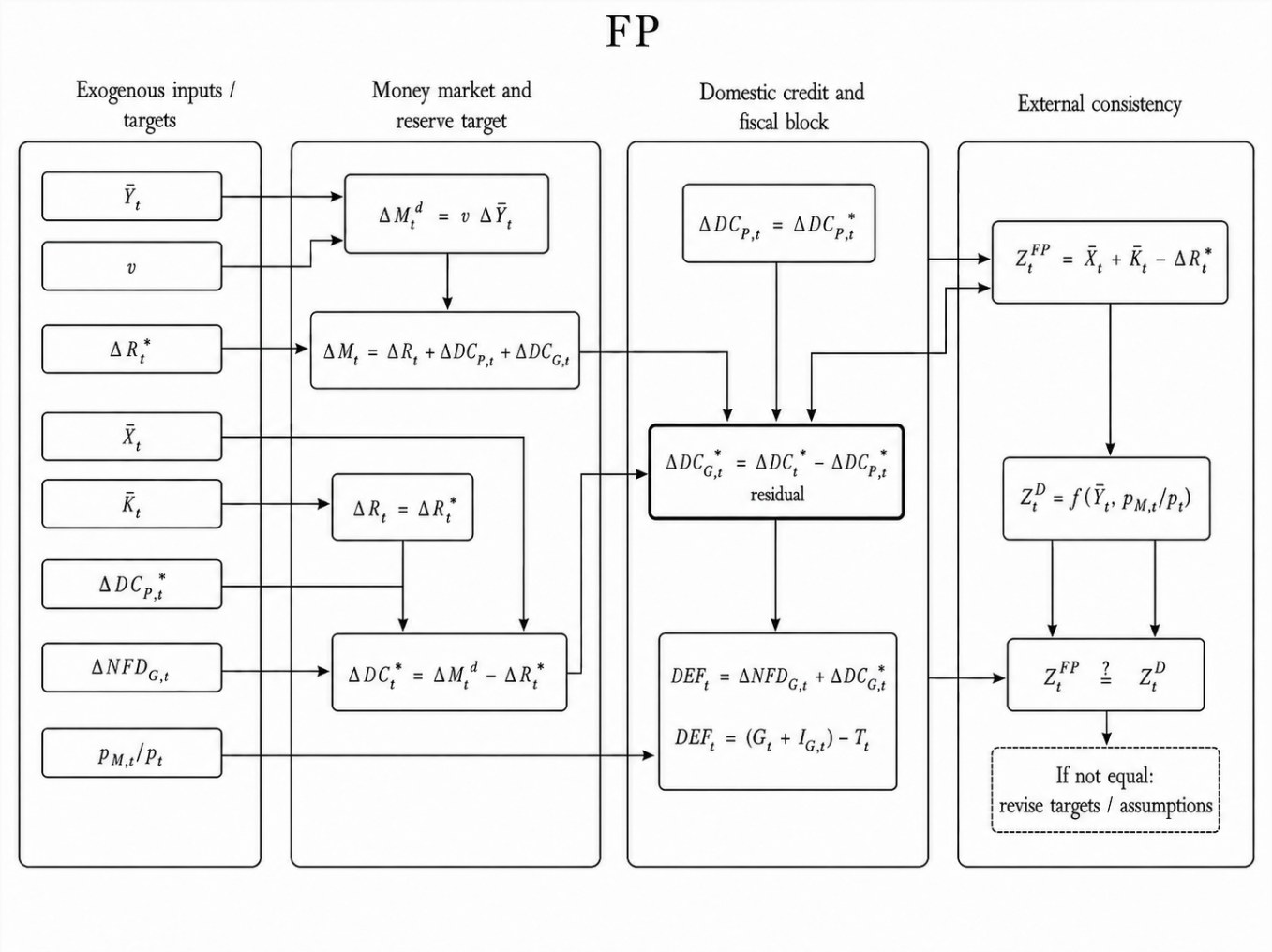
$$\underbrace{RESBAL_t,\ NETFSY_t,\ NETCTR_t}_{\text{Current external transactions}} \quad \Longrightarrow \quad \underbrace{CURBAL_t}_{\text{Current-account balance}}$$

$$\underbrace{CURBAL_t,\ AMTGAP_t,\ LTCAPF_t,\ OTHCAP_t,\ CHGRES_t}_{\text{External financing conditions}} \quad \Longrightarrow \quad \underbrace{GAPF}_{\text{Required foreign}}$$

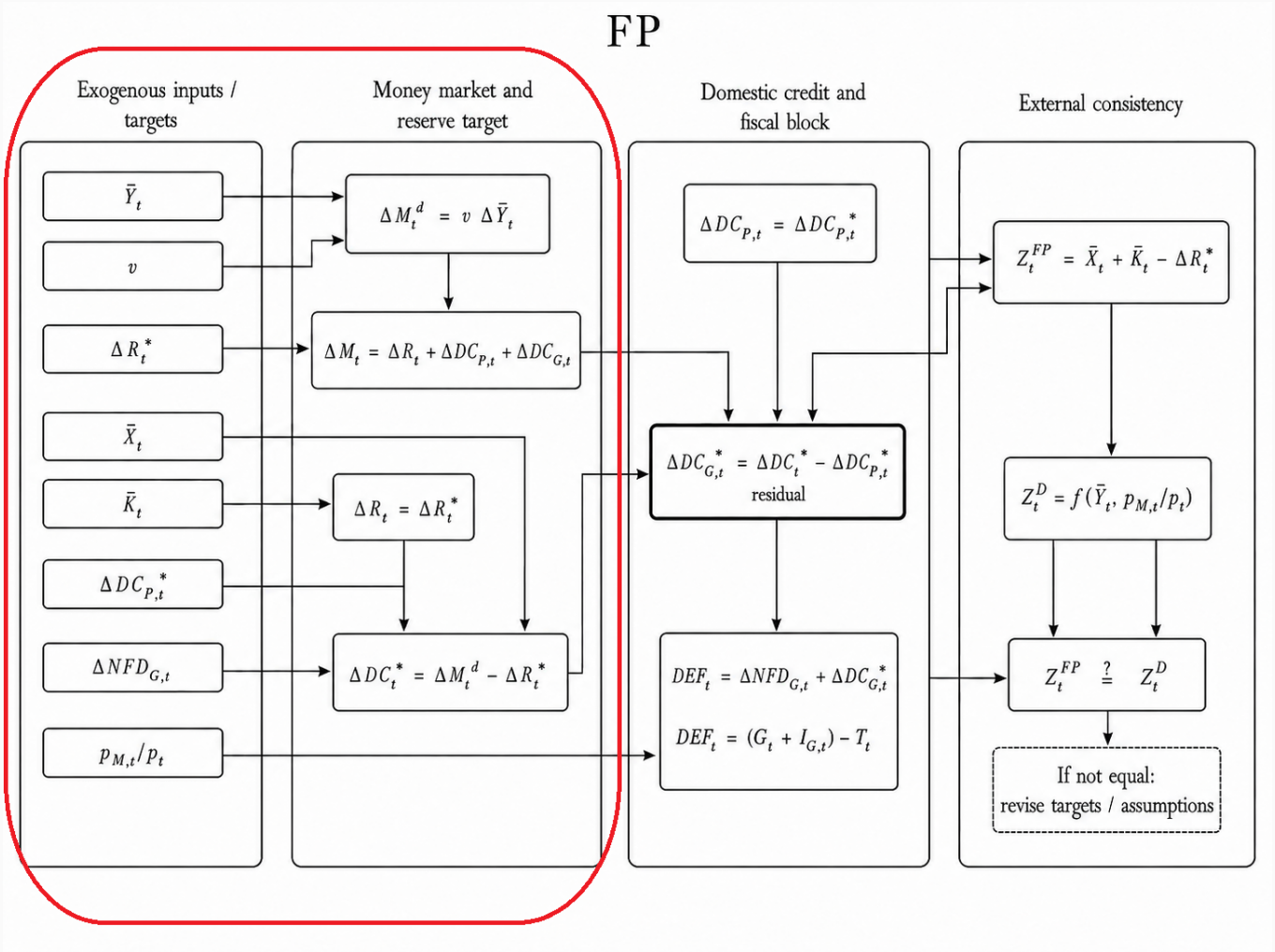
$$GAPFIL_t \quad \Longrightarrow \quad DODGAP_t \quad \Longrightarrow \quad INTGAP_{t+1}$$

FP

FP model


FP
Exogenous inputs / targets
Money market and reserve target
Domestic credit and fiscal block
External consistency
$\bar{Y}_t$
v
ΔR_t^*
$\bar{X}_t$
$\bar{K}_t$
$\Delta DC_{P,t}^*$
$\Delta NFD_{G,t}$
$p_{M,t}/p_t$
$\Delta M_t^d = v\,\Delta\bar{Y}_t$
$\Delta M_t = \Delta R_t + \Delta DC_{P,t} + \Delta DC_{G,t}$
$\Delta R_t = \Delta R_t^*$
$\Delta DC_t^* = \Delta M_t^d - \Delta R_t^*$
$\Delta DC_{P,t} = \Delta DC_{P,t}^*$
$\Delta DC_{G,t}^* = \Delta DC_t^* - \Delta DC_{P,t}^*$
residual
$DEF_t = \Delta NFD_{G,t} + \Delta DC_{G,t}^*$
$DEF_t = (G_t + I_{G,t}) - T_t$
$Z_t^{FP} = \bar{X}_t + \bar{K}_t - \Delta R_t^*$
$Z_t^D = f(\bar{Y}_t, p_{M,t}/p_t)$
$Z_t^{FP} \stackrel{?}{=} Z_t^D$
If not equal:
revise targets / assumptions

FP - Reserve


FP
Exogenous inputs / targets
Money market and reserve target
Domestic credit and fiscal block
External consistency
$\bar{Y}_t$
v
ΔR_t^*
$\bar{X}_t$
$\bar{K}_t$
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If not equal:
revise targets / assumptions

FP: Nominal GDP path

$$\underbrace{Y_t^N}_{\text{Nominal GDP}} = \underbrace{P_t}_{\text{Price level}} \underbrace{\bar{Y}_t}_{\text{Real GDP path}}$$

FP: Money demand

$$\underbrace{M_t^d}_{\text{Money demand}} = \underbrace{v}_{\text{Money-demand coefficient}} \underbrace{Y_t^N}_{\text{Nominal GDP}}$$

$$\Longrightarrow$$

$$\underbrace{\Delta M_t^d}_{\text{Required money growth}} = v \Delta Y_t^N$$

FP: Sources of money supply

$$\underbrace{\Delta M_t}_{\text{Change in money supply}} = \underbrace{\Delta R_t}_{\text{Reserve accumulation}} + \underbrace{\Delta DC_t}_{\text{Domestic-credit expansion}}$$

FP: Money-market equilibrium

$$\underbrace{M_t^s}_{\text{Money supply}} = \underbrace{M_t^d}_{\text{Money demand}}$$

$$\Longrightarrow$$

$$\underbrace{\Delta M_t^d}_{\text{Required money growth}} = \underbrace{\Delta R_t}_{\text{Reserve accumulation}} + \underbrace{\Delta DC_t}_{\text{Domestic-credit expansion}}$$

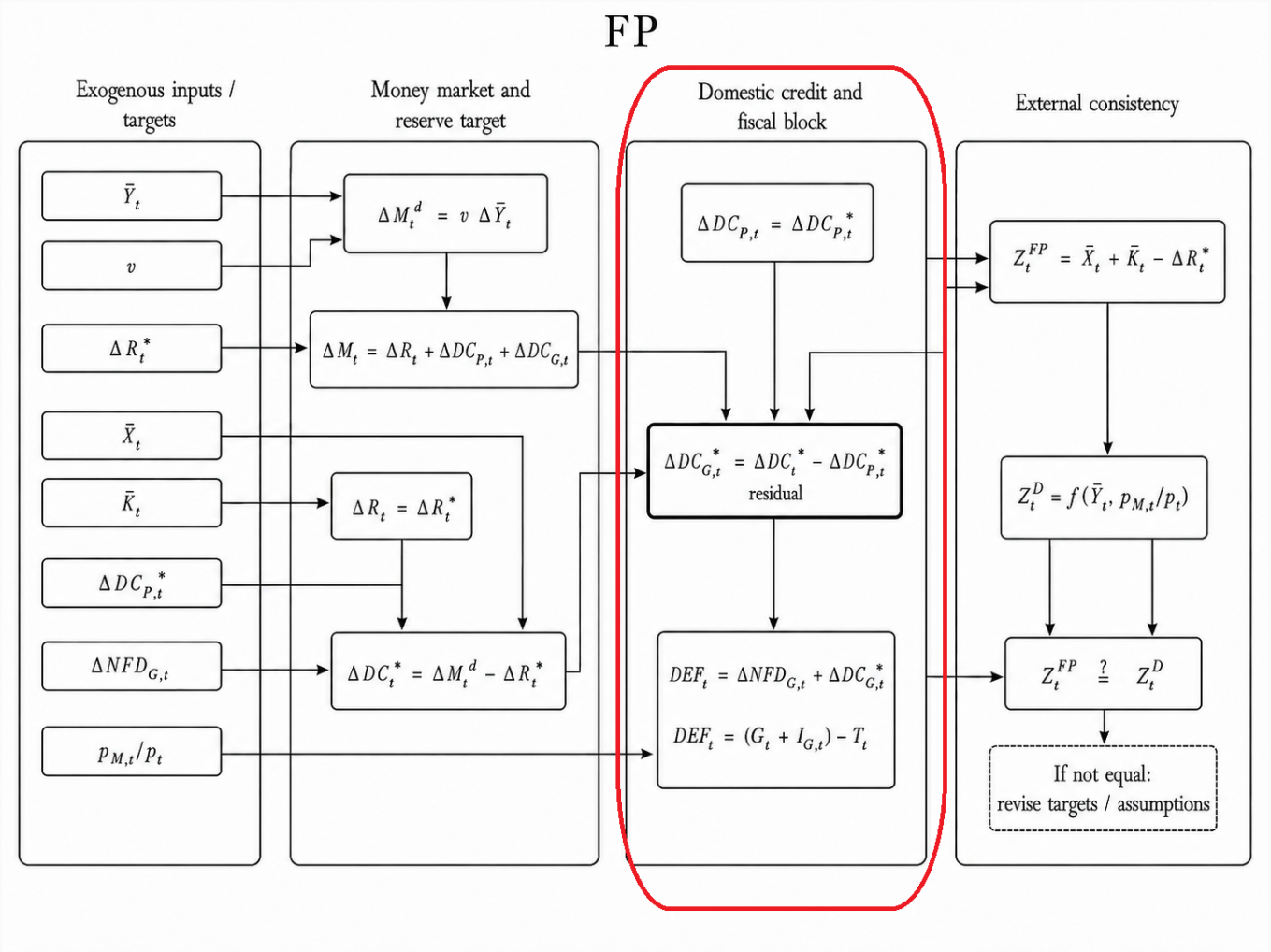
FP: Reserve target and domestic-credit ceiling

$$\underbrace{\Delta R_t}_{\text{Reserve accumulation}} = \underbrace{\Delta R_t^*}_{\text{Reserve target}}$$

$$\underbrace{\Delta M_t^d}_{\text{Required money growth}} = \underbrace{\Delta R_t^*}_{\text{Reserve target}} + \underbrace{\Delta DC_t^*}_{\text{Domestic-credit ceiling}}$$

$$\boxed{\underbrace{\Delta DC_t^*}_{\text{Domestic-credit ceiling}} = \underbrace{\Delta M_t^d}_{\text{Required money growth}} - \underbrace{\Delta R_t^*}_{\text{Reserve target}}}$$

FP - Domestic


FP
Exogenous inputs / targets
Money market and reserve target
Domestic credit and fiscal block
External consistency
$\bar{Y}_t$
v
ΔR_t^*
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$Z_t^{FP} \stackrel{?}{=} Z_t^D$
If not equal:
revise targets / assumptions

FP: Private-credit target

$$\underbrace{\Delta DC_{P,t}}_{\text{Private domestic credit}} = \underbrace{\Delta DC^{*}_{P,t}}_{\text{Private-credit target}}$$

FP: Government domestic credit

$$\underbrace{\Delta DC^*_{G,t}}_{\text{Government domestic credit}} = \underbrace{\Delta DC^*_t}_{\text{Total credit ceiling}} - \underbrace{\Delta DC^*_{P,t}}_{\text{Private-credit target}}$$

$\Delta DC^*_{G,t} =$ credit remaining for government

FP: Government deficit

$$\underbrace{DEF_t}_{\text{Government deficit}} = \underbrace{G_t}_{\text{Government expenditure}} + \underbrace{I_{G,t}}_{\text{Government investment}} - \underbrace{T_t}_{\text{Tax revenue}}$$

FP: Financing the government deficit

$$\underbrace{DEF_t}_{\text{Government deficit}} = \underbrace{\Delta NFD_{G,t}}_{\text{Government foreign borrowing}} + \underbrace{\Delta DC^*_{G,t}}_{\text{Government domestic credit}}$$

$$G_t + I_{G,t} - T_t = \Delta NFD_{G,t} + \Delta DC^*_{G,t}$$

FP: Domestic credit and fiscal block — summary

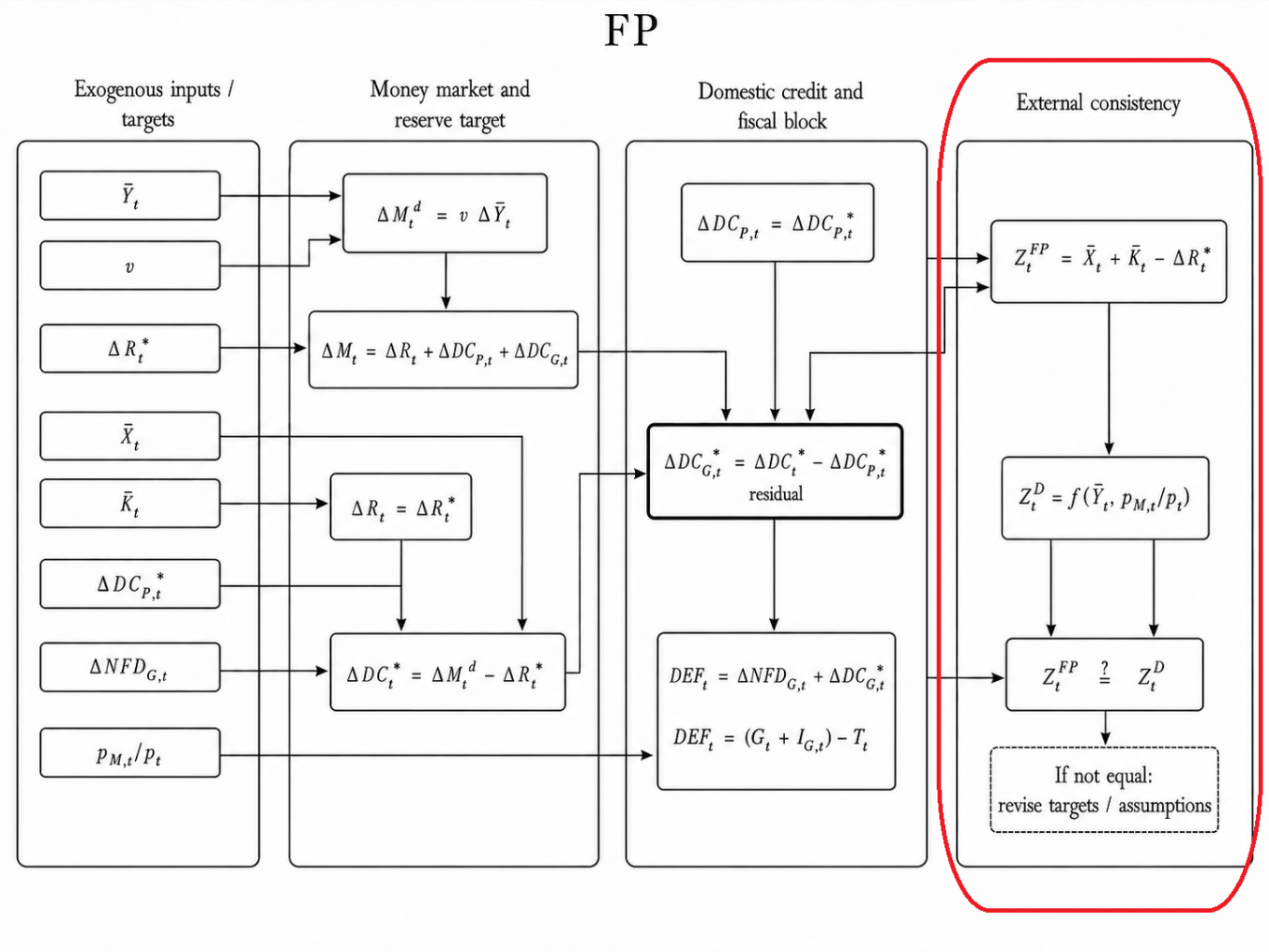
$$\underbrace{\Delta DC_t^*}_{\text{Credit ceiling}} - \underbrace{\Delta DC_{P,t}^*}_{\text{Private-credit target}}$$

$$\Longrightarrow$$

$$\underbrace{\Delta DC_{G,t}^*}_{\text{Government domestic credit}}$$

$$\underbrace{\Delta DC_{G,t}^*}_{\text{Domestic financing}} + \underbrace{\Delta NFD_{G,t}}_{\text{Foreign financing}}$$

FP - Foreign


FP
Exogenous inputs / targets
Money market and reserve target
Domestic credit and fiscal block
External consistency
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If not equal:
revise targets / assumptions

FP: External financing

$$\underbrace{\bar{K}_t}_{\text{Total external financing}} = \underbrace{\Delta NFD_{P,t}}_{\text{Private foreign financing}} + \underbrace{\Delta NFD_{G,t}}_{\text{Government foreign financing}}$$

FP: Balance-of-payments identity

$$\underbrace{\Delta R_t}_{\text{Reserve accumulation}} = \underbrace{\bar{X}_t}_{\text{Exports}} - \underbrace{Z_t}_{\text{Imports}} + \underbrace{\bar{K}_t}_{\text{External financing}}$$

$$\underbrace{\bar{X}_t + \bar{K}_t}_{\text{Foreign exchange inflows}} = \underbrace{Z_t + \Delta R_t}_{\text{Foreign exchange uses}}$$

FP: Imports that can be financed

$$\underbrace{\Delta R_t}_{\text{Reserve accumulation}} = \underbrace{\Delta R_t^*}_{\text{Reserve target}}$$

$$\Longrightarrow$$

$$\underbrace{Z_t^{FP}}_{\text{Imports that can be financed}} = \underbrace{\bar{X}_t}_{\text{Exports}} + \underbrace{\bar{K}_t}_{\text{External financing}} - \underbrace{\Delta R_t^*}_{\text{Reserve target}}$$

FP: Import demand

$$\underbrace{Z_t^D}_{\text{Import demand}} = \underbrace{\mu_t}_{\text{Marginal propensity to import}} \underbrace{Y_t^N}_{\text{Nominal GDP}}$$

$$\Longrightarrow$$

$$\underbrace{\Delta Z_t^D}_{\text{Change in import demand}} = \underbrace{\mu_t}_{\text{Marginal propensity to import}} \underbrace{\Delta Y_t^N}_{\text{Nominal GDP growth}}$$

FP: External-consistency check

$$\underbrace{Z_t^{FP}}_{\text{Imports that can be financed}} \stackrel{?}{=} \underbrace{Z_t^{D}}_{\text{Import demand}}$$

$Z_t^D > Z_t^{FP} \implies$ Import demand exceeds available foreign exchange

$Z_t^D < Z_t^{FP} \implies$ Financeable imports exceed import demand

FP: External consistency — summary

$$\underbrace{\bar{X}_t,\ \bar{K}_t,\ \Delta R_t^*}_{\text{External programme assumptions}} \quad \Longrightarrow \quad \underbrace{Z_t^{FP}}_{\text{Imports that can be financed}}$$

$$\underbrace{\mu_t,\ Y_t^N}_{\text{Import-demand conditions}} \quad \Longrightarrow \quad \underbrace{Z_t^D}_{\text{Import demand}}$$

$$\boxed{Z_t^{FP} \stackrel{?}{=} Z_t^D}$$

If not equal: revise programme assumptions and solve again.

Merged Model

From RMSM and FP to the Merged Model

RMSM

$GDP,\ X \longrightarrow I,\ M$
$\longrightarrow$ real resource/trade outcome

Growth and export paths determine the implied investment, imports, and trade outcome.

FP

government finance $\longrightarrow$ money/credit
$\longrightarrow$ external finance

Fiscal, monetary, and external accounts must remain mutually consistent.

$$\boxed{\text{Merged Model}} = \underbrace{\text{RMSM real-side core}}_{\text{growth, investment, imports, trade}} + \underbrace{\text{FP financial core}}_{\text{fiscal, money, credit, financing}}$$

The merge changes the closures

$$\underbrace{C_t}_{\text{RMSM residual}} \Longrightarrow \underbrace{C_{P,t}}_{\text{determined from private disposable income}} + \underbrace{C_{G,t}}_{\text{Merged Model focal variable}}$$

$$P_{D,t} C_{P,t} = (1 - b_t) GDY_t$$

$$\underbrace{GAPFIL_t}_{\text{RMSM residual financing}} \Longrightarrow \underbrace{NFD_{P,t}}_{\text{Merged Model focal variable}}$$

The residuals become institutionally specific.

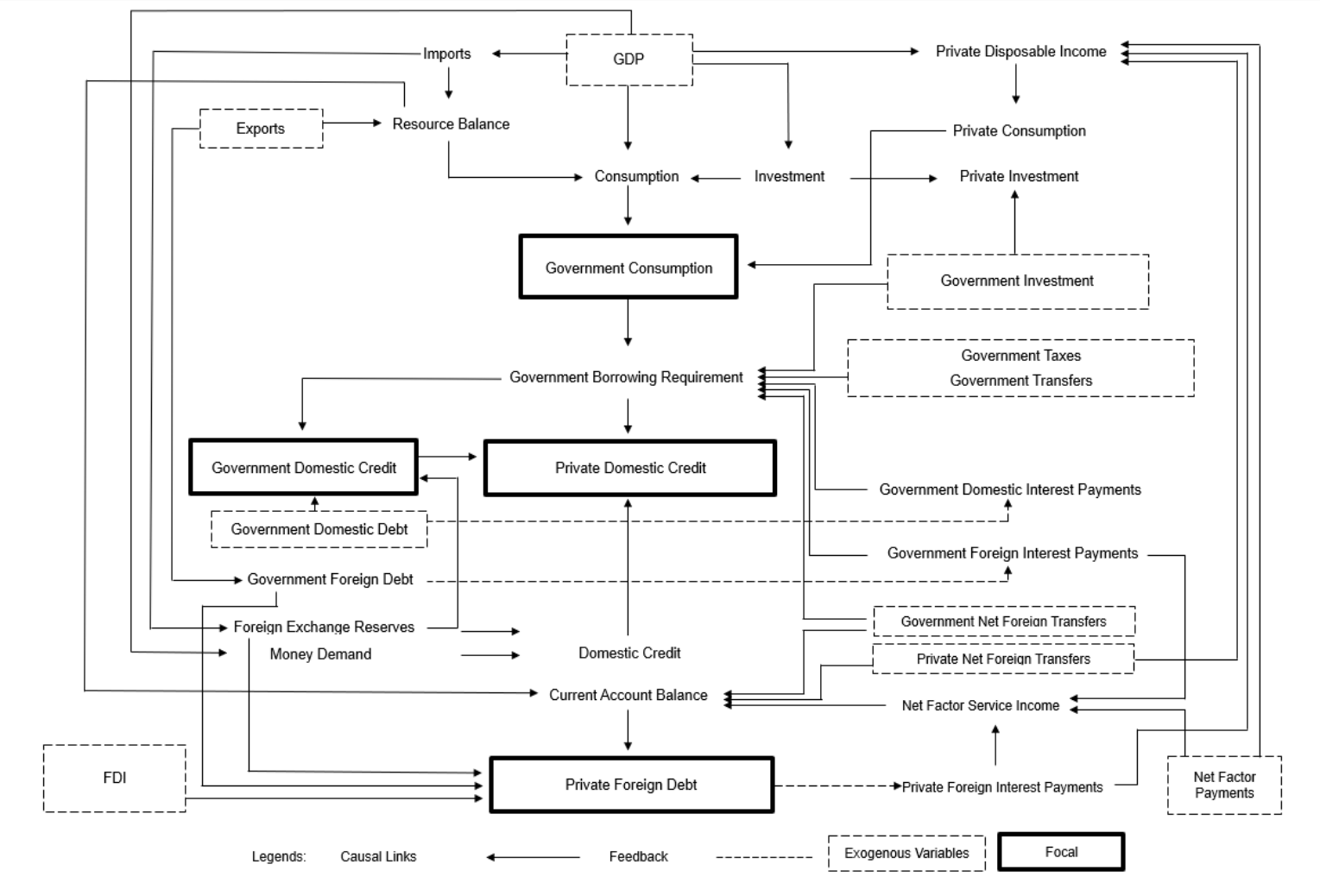
Merged Model: How the parts connect

$$\underbrace{GDP_t,\ X_t \Longrightarrow I_t,\ M_t}_{\text{RMSM real-side logic}} \Longrightarrow \underbrace{C_{G,t}}_{\text{first focal variable}}$$

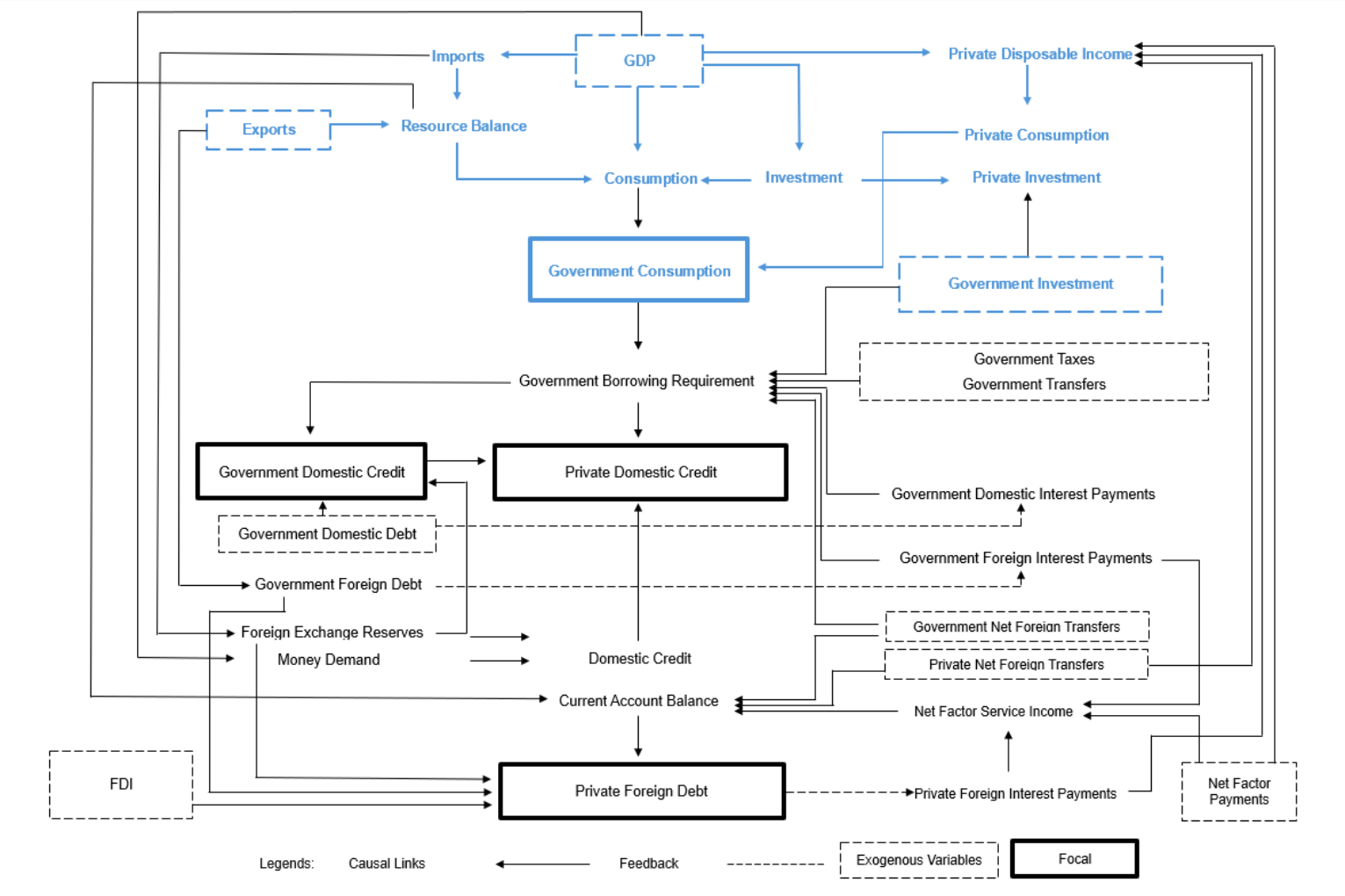
$$\underbrace{C_{G,t} \Longrightarrow BR_{G,t} \Longrightarrow DC_{G,t}}_{\text{government budget}}$$

$$\underbrace{P_t GDP_t,\ \Delta(E_t R_t),\ DC_{G,t} \Longrightarrow DC_{P,t}}_{\text{money market}}$$

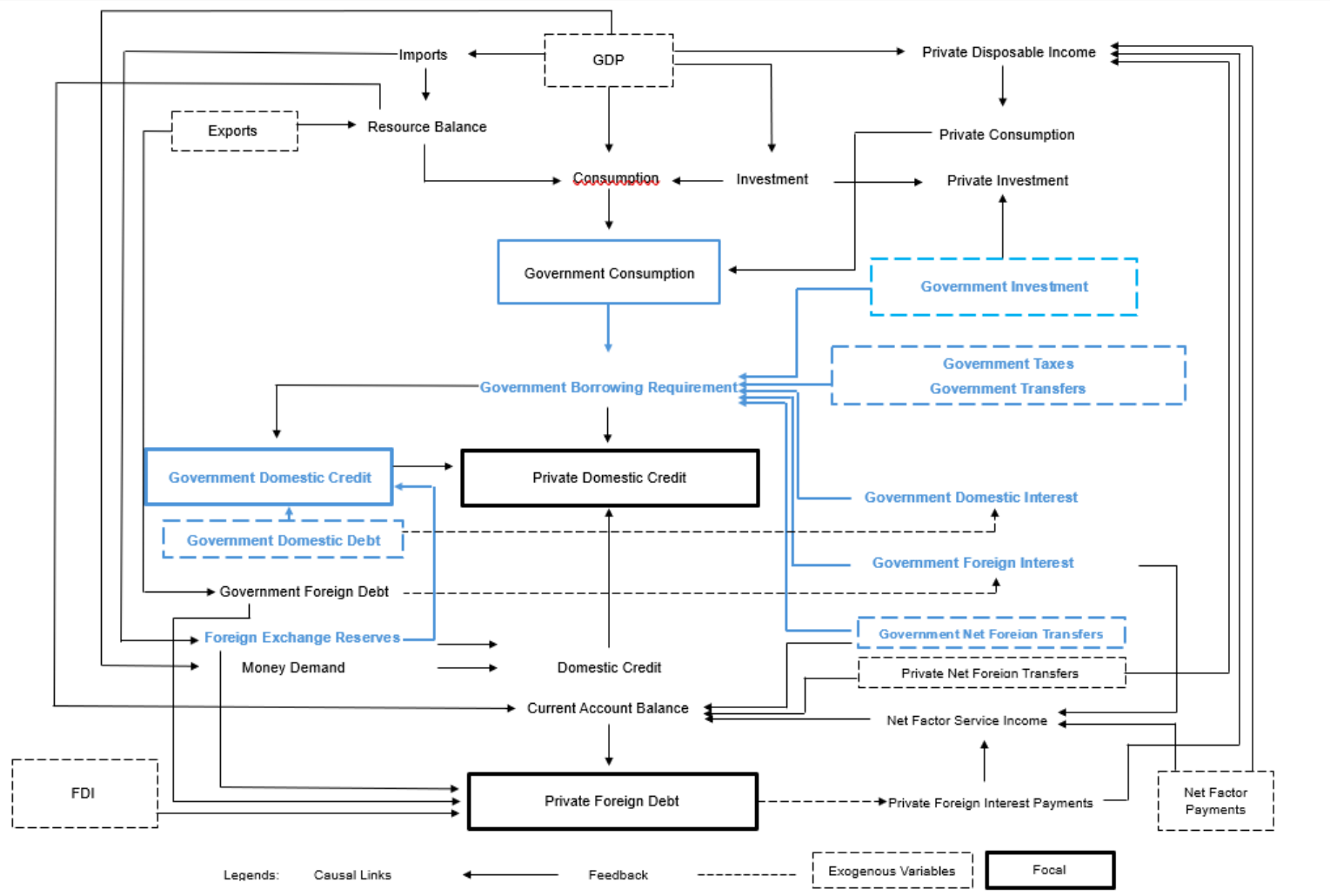
Merged Model



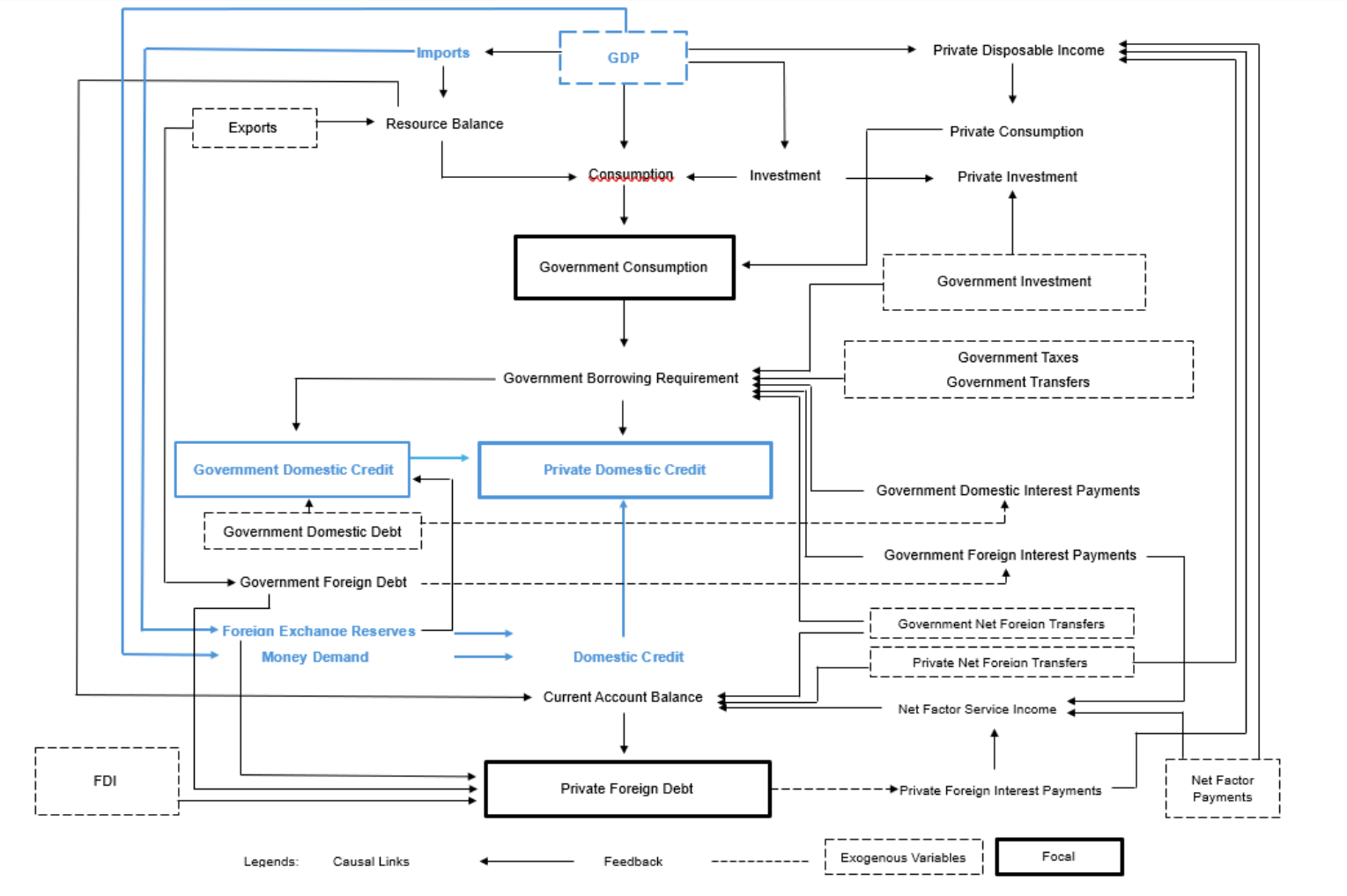
Merged Model: Goods market and private sector



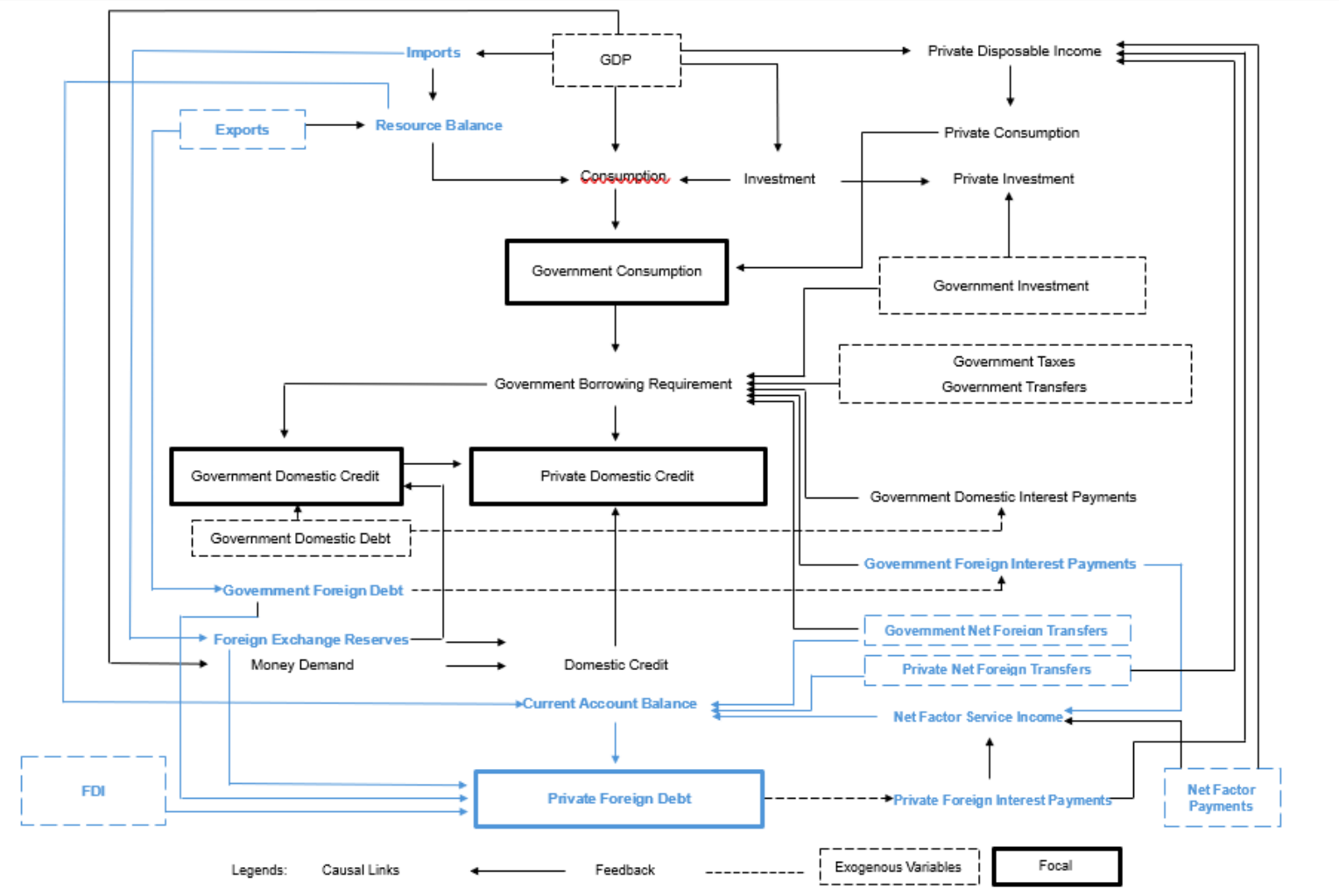
Merged Model: Government budget



Merged Model: Money market



Merged Model: Balance of payments



Merged Model: Focal variables I

Government consumption

$C_{G,t}$ Does the scenario leave a plausible path for government consumption?

Government domestic credit

$DC_{G,t}$ Is the required government credit expansion plausible?

Merged Model: Focal variables II

Private domestic credit

$DC_{P,t}$ Is the required private credit expansion plausible?

Private foreign debt

$NFD_{P,t}$ Is the required private foreign borrowing plausible?

Preparation for tomorrow

Before tomorrow, please set up your programming environment:

- Download and install **Julia**
- Download and install **VSCode**
- Open VSCode and confirm that Julia runs in the integrated terminal

Thank you for today

See you tomorrow.